

User Manual

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1. Introduction

OctoPus (Octave + Jianpu) is a desktop editor for numbered music notation. Write a score as JPS (Jianpu Script) text, review a live SVG score, compare it with an original image or PDF, and export finished pages. A .jps file describes musical content; page settings control its presentation. This guide describes version 0.1.0, updated 1 October 2026.

The program works locally for editing, rendering and supported transcription once installed. Help links and update checks need a browser or network connection. This manual works offline; its English and Chinese versions share the same topics and examples. Use the language tabs, or download the matching PDF.

2. Capabilities, limits and roadmap

Available now	Scope
JPS editing	New/open/save, undo/redo, clipboard, find/replace, source formatting and bundled samples.
Live SVG preview	Source/score correspondence, page navigation, zoom and page layout controls.
Reference transcription	PNG/JPG/JPEG/PDF input, original display and provisional JPS drafts for review.
Export	SVG, PDF and JPEG; locally packaged worker and bundled font substitutions.
Languages	English/Chinese interface and user manual.

Current limits. Transcription is not guaranteed to recover a complete score. Check notes, accidentals, octave dots, durations, lyrics, slurs, grace notes, endings and page order. General running-header/footer/page-number filtering is still incomplete. Complex layout and cross-page spans need manual review. There is no desktop playback/MIDI workflow, LilyPond exporter or MusicXML import/export UI. Staff-notation recognition is planned, not a supported replacement for Jianpu input. Automatic update installation is not provided. Native macOS packaging is deferred; Windows/macOS host acceptance is separate from Linux testing.

- Planned: improve transcription quality, including difficult marks and repeated page furniture.
- Planned: LilyPond source export and MusicXML export/import.
- Planned: import and recognize staff-notation sheet music as JPS.
- Plans indicate intended direction, not a release date or a current capability.

3. Your first score

- Start OctoPus and choose **New**. Enter a title, choose the basic score fields and confirm.
- Replace the source with the example below, or open a comparable score from **Samples Library**.
- Wait for the SVG preview to finish rendering. Click a note to locate its source; edit the text and review the result.
- Use **Page Settings** for margins/fonts, **Save As** for a .jps file, then **Export** for the finished score.

```
V: 1.0
B: First score
Z: Practice example
D: C
P: 4/4
J: 90

Q: 1 2 3 4 | 5 6 7 1' ||
C: 春风吹来花儿开了
```

This example has two four-note bars; the final note is one octave above the base register. The eight Chinese lyric characters map to the eight notes.

4. The workspace and UI

The top row provides New, Open, Save, Save As, Export, Close, Help, Import PNG/JPG/PDF, Samples Library, Page Settings, Language and Exit. The logo icon and text open the project home. The second row selects Mode/Layout, adjusts split sizes and focuses a pane.

Pane	Purpose
JPS source	Edit text. Filename and Unsaved status sit immediately before the right-aligned Format Source button. Editing actions follow the JPS source heading.
SVG preview	Rendered score, fit/zoom/page controls and render diagnostics.
Original Image/PDF	Reference page, fit/zoom/rotate controls and Transcribe menu; visible in Transcription mode.

On a narrow pane, view controls become a **View** menu and source tools become **Edit**. SVG View includes Fit Page, Fit Width, Prev Page and Next Page; Original View includes fit and rotation plus PDF page navigation when applicable. Very narrow panes also expose zoom through View. This keeps actions on one heading row.

5. Files, saving and recovery

Open loads a .jps score. Both plain JPS text and compatible JSON-wrapped JPS documents are supported. The wrapper may contain source, page settings and existing metadata. **Save** writes the current document; a new document first needs a destination. **Save As** changes the destination. Changing page settings for plain text may require a wrapper to preserve them; heed the dialog notice.

Close closes the score; **Exit** closes the application. If changes are unsaved, choose Save, Discard or Cancel. Recovery can restore the last recoverable draft and managed references after interruption; it does not replace ordinary saved backups. Reference images/PDFs are separate from score source: retain their original files. Opening a sample creates a working document; choose your own save location before overwriting anything.

6. Editing JPS source

Use Undo/Redo, Cut/Copy/Paste, Find, Replace and Select All from the heading or compact Edit menu. Find opens the search UI; Replace exposes replacement controls. Standard editor keyboard shortcuts remain available. Clipboard availability depends on the operating system. **Format Source** spaces musical tokens while retaining notation content; it cannot repair musical mistakes or validate a transcription. Quoted annotations and attached grace groups need careful review after edits.

Prefer ASCII punctuation for syntax: colon :, quote ", comma , apostrophe ' and slash /. Use full-line # comments to explain the source. Do not paste curly quotes or a full-width colon as syntax. Keep attached modifiers next to their note. Preserve a copy before a large replacement and use Undo when needed.

7. Preview, cursor and navigation

The preview updates as source changes. Clicking or moving the source cursor highlights the matching score position and brings that row toward the preview center. Clicking a preview note or lyric selects its source without automatically moving the preview. Direct source clicks/keyboard movement resume following. Only uniquely mapped elements are selectable; some decorative or temporary-voice elements may not have a precise source target.

Use -/+ to zoom, Fit Page for the whole page and Fit Width for width-based reading. Page arrows or View menu items navigate rendered pages. Refresh requests a new render. Click diagnostics below the preview to jump to the related source. A previous preview can remain visible during rendering; wait for completion and resolve relevant warnings before exporting.

8. Original images/PDFs and transcription

Choose Import PNG/JPG/PDF or drop **one** PNG, JPG/JPEG or PDF into Original Image/PDF. An existing reference prompts Replace Reference or Cancel. Import is blocked while another import, replacement decision, recovery, save, transcription or conflicting dialog is active. Unsupported/multiple dropped files are rejected without import.

Drag the displayed original to pan; use fit, zoom and Rotate 90°. Rotation changes the viewing orientation, not the original file. A PDF has page navigation. The pane close control returns to Normal mode; the managed reference is retained. Input limits include 100 MiB per image/PDF, at most 200 PDF/reference pages and 40 million pixels per decoded image/PDF page. Damaged, encrypted or unsupported files may fail; reduce the input if the program reports a size or page-limit error.

- Transcribe New: generates a provisional new score. Handle any unsaved-score prompt before replacing the working score.
- Transcribe Append: adds recognized music to the current source, using a page break boundary when needed and omitting repeated draft headers for a nonempty score. Undo can reverse the insertion.
- A PDF is transcribed as a document, not only its displayed page. Review the warnings and each page against the original; collapse warnings when you need editing space.
- Correct uncertain recognition manually. Check octave dots, beams, dotted durations, accidentals, grace notes, slurs, endings and lyrics before saving/exporting.

9. Modes, layouts and focus

Mode/layout	Use
Normal N1	JPS source and preview side by side.
Normal N2	JPS source and preview stacked.
Transcription T1	Original, source and preview in three columns.
Transcription T2	Original and preview above the source editor.

Use the width/height sliders to adjust splits. Focus Original, Focus Editor or Focus Preview temporarily expands one pane; Exit Focus restores the layout. Controls adapt as the available width changes. F11 toggles the window maximized state; the toolbar has no separate Maximize button.

10. Page settings and fonts

Page Settings groups paper (A4/A5), margins and body offset; title/lyric fonts and note style; title/subtitle/lyric sizes; music-to-lyric, lyric-to-lyric, lyric-to-music and voice spacing; and slur shape (Automatic, Arc, Flat top). Sizes/margins are renderer units, not millimeters. **Apply** updates the score while keeping the dialog open; **OK** applies and closes; **Cancel** discards unapplied draft changes. Already applied changes are not reverted by Cancel. Save the score to retain changes.

Role	Windows/Linux preference	macOS preference	Bundled fallback
HeiTi-1 (default)	Microsoft YaHei	PingFang SC	MiSans Regular
HeiTi-2	SimHei	Heiti SC	LXGW Neo XiHei
SongTi	SimSun	Songti SC	SimZhiSong
KaiTi	KaiTi	Kaiti SC	LXGW WenKai Regular
FangSong	FangSong	STFangsong	Zhuque Fangsong Regular

Each installed family is preferred independently, including in production portable builds. Linux needs an exact fontconfig match. Missing families use bundled substitutions. Bundled text with missing characters tries Noto Sans SC, then Noto Serif SC for coverage; neither guarantees every Unicode character. Latin Arial requests use Liberation Sans (Regular/Italic/Bold/Bold Italic). Note styles are Regular, Italic and Bold. Saved HeiTi maps to HeiTi-2. Font substitutions can change spacing. Microsoft fonts are not redistributed. MiSans has its own Xiaomi license; other bundled notices include OFL and IPA licenses.

11. Export and portable use

Choose Export and select SVG, PDF, JPG (96 DPI) or JPG (300 DPI), then choose the output destination. PDF produces one multipage document; SVG/JPG produce page files. 300 DPI is useful for print-quality raster output. The disabled LilyPond item is a planned feature. Save the JPS as the editable master before sharing rendered files. PDF embeds fonts, JPEG stores pixels and SVG preserves vector content, but external SVG viewers need matching text fonts. Inspect the result in its destination viewer. The separate Python ReportLab PDF path cannot embed CFF-outline system fonts and uses the role fallback there; this is not a blanket limitation of native desktop export.

Keep the entire portable folder together, including engine, examples and docs. On Linux run `dist/x86_64-unknown-linux-gnu/portable/usr/bin/octopus`; Windows uses `portable/octopus.exe`. The worker is bundled, so no separate Python is required. Linux still requires GTK/WebKit runtime libraries and Poppler; Windows requires WebView2. Portable does not mean every host dependency is embedded.

12. Keyboard and Help

Shortcut	Action
Ctrl/Cmd+N	New score
Ctrl/Cmd+O	Open score
Ctrl/Cmd+S / Shift+S	Save / Save As
Ctrl/Cmd+Z	Undo
Ctrl/Cmd+Y or Shift+Z	Redo
Ctrl/Cmd+E	Insert the editor' s last inserted content when available
Ctrl/Cmd+X / C / V / A	Cut / Copy / Paste / Select All in the editor
Ctrl/Cmd+F	Editor search
F11	Toggle window maximization
Arrow / Home / End / Escape	Navigate menus; Escape dismisses. Language tabs also support arrow/Home/End.

Some shortcuts are inactive while a modal dialog is open or a form control has focus. Help offers User Manual, Report Issues, Submit Requests, Check for Update and About OctoPus. Issues open the project issue page; Requests opens the pull-request page. Check for Update compares the latest stable GitHub release and reports unavailable information, unfamiliar tags or network errors separately; it does not install updates. About and the logo link to the project home.

13. JPS structure and conventions

JPS is ordinary text, normally stored with extension .jps. A score has a descriptive header followed by music Q: and lyrics C:. One Q line represents a printed music row; following C lines belong to that row. Numbered prefixes identify voices. Blank lines improve source readability. A whole line beginning with # is a comment; inside a Q line, # is an accidental, not a comment.

The original specification calls V, B, D and P required fields. Include them for a complete portable score even though OctoPus tolerates incomplete drafts. Use uppercase header/prefix letters and lowercase literal [fenye]. Preserve the header/body distinction; repeated B/Z fields serve specific layout purposes. Unknown syntax may be preserved with diagnostics rather than rendered as intended.

14. Descriptive header

Field	Example value	Meaning
V	1.0	Script version / 脚本版本
B	Practice	Title; repeats become subtitles / 标题，重复成为副标题
Z	Name - composer	Author/credit; multiple right-aligned lines / 作者，可多行右对齐
D	E\$	Key; A-G with # sharp or \$ flat / 调号，A-G 加 # 或 \$
P	4/4	Meter; alternatives may follow / 拍号，可附其他拍号
J	90	Tempo number and/or descriptive text / 速度数值及/或文字
X	10	Score sequence number / 乐谱序号

V: 1.0
B: Main title
B: Subtitle
Z: Writer - lyrics
Z: Composer - music
D: F#
P: 4/4 (2/4 1/4)
J: 90 Allegro
X: 10

Use a capital A-G key letter: C, F#, E\$. Multiple B lines put subtitle text under the first title; multiple Z lines place credits at the upper right. P uses slash-separated numerator/denominator; spaces and parentheses can describe auxiliary meters. J may be numeric beats per minute, words, or both. X is metadata, not a voice number. Header values describe the score; a tempo field does not enable playback.

15. Notes, register and duration

Syntax	Meaning
1-7	Scale degrees / 简谱音级
0	Rest / 休止符
8	Invisible rest with layout space / 占位隐藏休止符
9	Rhythm X note / 节奏 X 音符
1' / 1''	One/two octaves above / 高一/两个八度
1, / 1,,	One/two octaves below / 低一/两个八度
1 - -	Quarter note plus two extension beats / 音符加两拍增时线
1 / 1 // 1 ///	Eighth / sixteenth / thirty-second / 八分、十六分、三十二分
1. / 1..	Single/double dot / 单附点、双附点

A bare digit is the basic quarter-note unit in the compatibility notation. Each attached slash halves its duration. A dot adds half the undotted duration; two dots add half plus a quarter (1.75 times the original). Extension hyphens add the Jianpu continuation strokes. They do not each need a lyric skip. Add octave commas/apostrophes after the digit; avoid spaces between the note and its suffix.

```
Q: 12' 3,, 0 | 1/2/3//4//5/6 7 | 6. 7/1' - ||
```

16. Custom beat grouping

The reference notation uses quarter-note-based automatic beam grouping, which can be unsuitable for 6/8 or other compound/special grouping. Use `~` to join the adjacent notes in one group and `^` to force a split. These control visual beam grouping, not a new pitch or lyric syllable. Compare against the intended beat pattern rather than relying only on the P field.

```
P: 6/8  
Q: 1/~2/~3/^4/~5/~6/ |
```

17. Accidentals and grace notes

Write sharp #, flat \$ or natural = after the host digit, for example 4#, 7\$, 3=. Grace notes are attached square-bracket groups after their host: `1[23]` displays preceding grace notes; `1[h23]` displays following grace notes. Inside a group, commas/apostrophes, slashes and accidentals still apply. Grace digits default to eighth notes; a slash makes a sixteenth. A bracket attached to a note is not an ending bracket on a barline.

```
Q: 4# 7$ 3= 1[2/3'] | 5[h6,7/] - ||
```

18. Dynamics and hairpins

Attach an ampersand code to a note, such as `3&mp`. The reference terms are pp, p, mp, mf, f, ff, fff, cresc, dim, sf, fp, sfp, a tempo and rit. Use `&a tempo` for the two-word term. Hairpins start with `<` (crescendo) or `>` (decrescendo) and end with `!`. Add `+` after the opening marker to move it upward when it overlaps another span. These are printed instructions; the desktop app does not interpret them as audio playback controls.

```
Q: 1&p<2 3 4! | 5>+ 4 3 2! | 1&a tempo - - ||
```

Code	Instruction
<code>&pp</code>	Very soft
<code>&p</code>	Soft
<code>&mp</code>	Moderately soft
<code>&mf</code>	Moderately loud
<code>&f</code>	Loud
<code>&ff</code>	Very loud
<code>&fff</code>	Extremely loud
<code>&cresc</code>	Crescendo
<code>&dim</code>	Diminuendo
<code>&sf</code>	Sudden accent
<code>&fp</code>	Loud then immediately soft
<code>&sfp</code>	Accented then immediately soft
<code>&a tempo</code>	Return to original tempo
<code>&rit</code>	Gradually slower

19. Ornaments, parentheses and annotations

Code after a note	Meaning
&yc	Fermata / 延长记号
&bc	Tenuto / 保持音
&zy	Accent / 重音
&dy	Staccato / 顿音
&hx	Breath / 呼吸
­ / &xhy	Upward / downward slide / 上、下滑音
&sby / &xby	Upper / lower mordent / 上、下波音
&cy	Trill / 颤音
&sby+ / &xby+ / &cy+	Variant ornament forms / 装饰变体
&zkh / &ykh	Left / right accompaniment parenthesis / 伴奏左右括弧

These short codes come from the notation reference, often from Chinese name initials. Attach them to the correct note. The + suffix selects the illustrated alternative for mordents/trills. Add a quoted annotation after a note to print text above it. Accompaniment parentheses &zkh and &ykh are also placed after the host digit even when the drawn parenthesis appears before the music.

```
Q: 3&yc 3&bc 3&zy 3&dy | 3&hx 3&shy 3&xhy 3&cy |  
Q: 1&zkh 2 3 4&ykh | 5"sing gently" - - - ||
```

20. Barlines and repeat instructions

Syntax	Appearance/use
	Single / 普通小节线
	Thin-left/thick-right final line / 左细右粗终止线
/	Two thin lines / 双细线
:	Start repeat / 左反复
:	End repeat / 右反复
: :	End/start repeat / 两侧反复
/	Hidden, no layout width / 不显示、不占位
*	Hidden, reserves layout width / 不显示、占位

Barline-attached code	Meaning
&fine	Fine / 终止
&dc	D.C. / 从头反复
&ds	D.S. / 从记号反复
&ty	Segno / 大跳记号
&hs	Coda / 花记号

Repeat/navigation marks belong to a barline, not an ordinary note. Use `|/` at a row beginning when a barline-attached mark is needed without a visible barline. Use `|*` to reserve alignment space, especially with mixed single/multiple voices. These marks are printed; playback navigation is not provided.

21. Slurs and tuplets

Wrap music in parentheses for a slur: `(1 2 3)`. Spans can nest and continue across barlines or music rows on the same page. Open the span in the earlier Q row and close it in the later row of the same voice. Use `(y1 2 3)` for a tuplet; the group count is inferred from its notes. Do not use a capital Y. Cross-page slurs are not automatically continuous: split/reopen spans at manual page boundaries and inspect the result.

```
Q: (1 (2 3) 4 | 5 6
Q: 7 1') | (y1/2/3/) 4 5 6 ||
```

22. Endings and temporary meter

Ending brackets [and] are attached to barlines. Put their label in a quoted barline annotation. A start at the beginning of a row uses hidden |/. Add / after [for an open right edge; add + (or several +) to raise the bracket and avoid overlapping slurs. Ending spans may cross rows; keep their endpoints in the correct voice. The same brackets attached to a note mean grace notes, so placement matters.

```
Q: |/"1" 1 2 3 4:] |/"2" 5 6 7 1' ||]
Q: |/[+"1,2" 1 2 3 4| 5 6 7 1' ]]
```

For a meter change, put "p:3/4" after the barline. This prints a local time signature; inspect grouping separately and use ~ or ^ where needed.

```
Q: 1 2 3 4|"p:3/4" 5 6 7 |
```

23. Voices and temporary accompaniment

Use Q1, Q2, etc. for music and C1, C2, etc. for the corresponding lyrics. A voice name can follow its number in quotes. Keep successive voices rhythmically aligned; numbered music rows form a system, while lyrics attach to their matching voice. Single and multiple voices can be mixed. A mid-row voice brace is indicated by &sbfb; use hidden rest 8 and hidden barline |* to reserve earlier space in secondary voices.

```
Q1"Soprano": 1 2 3 4 |
C1: 春风吹来
Q2"Alto": 3, 4, 5, 6, |
C2: 花儿开了
```

Temporary accompaniment uses {bz ...} ; temporary multiple voices use {dsb ...} . The keyword must immediately follow { and be lowercase. These inline groups are positioned against the main music. Copy an appropriate bundled example and check duration/alignment instead of assuming arbitrary blocks will align automatically.

```
Q: 3 4 | {bz 1&zkh 2 3 4 | 5 6 7 1'&ykh} 5 6 7 1' |
Q: 3 4 | {dsb 1 2 3 4 | 5 6 7 1'} 5 6 7 1' |
```

A minimal mid-row brace pattern (inspect alignment in the preview):

```
Q1: 1 2 3 4 |&sbfb 5 6 7 1' |
Q2: 8 8 8 8 |* 3 4 5 6 |
```

24. Lyrics: Chinese, English and mixed text

A C line belongs to the preceding music row; use matching voice numbers for multipart scores. Chinese characters ordinarily map one per note. Punctuation is recognized without becoming a separate sung syllable. @ skips a note; extension - strokes do not each need @. Join two Chinese characters with ~ to fit one note. Several C lines after one Q row supply separate verses.

```
Q: 1 2 3 4 | 5 - 6 7 |  
C: 春风@来花~儿开了  
C: 第二段词也能写
```

Begin a lyric line with a quoted label such as "1.". Use underscore _ for a space inside that legacy lyric-label notation. English words/syllables are separated by /; do not map individual letters to notes. / can also make mixed Chinese/English boundaries explicit. Use @ for unsung notes and inspect melismas rather than treating ordinary spaces as dependable English syllable delimiters.

```
Q: 1 2 3 4 |  
C: "Verse_1" Sing/a/new/song  
C: "2." 春/风/@/song
```

25. Manual page breaks

Place the exact lowercase token [fenye] on its own line. It creates a new score page; it is not a note annotation. Keep the intended music systems and their lyrics together. Long source does not imply automatic musical reflow or automatic page-spanning slurs. Split/reopen slurs at the page boundary, inspect ending brackets and confirm that each exported page fits its margins.

```
Q: 1 2 3 4 ||  
C: 第一页词  
  
[fenye]  
  
Q: 5 6 7 1' ||  
C: 第二页词
```

26. Practical editing and review checklist

- Choose a bundled sample close to the score: ordinary melody, symbols/grace notes, multiple voices or lyrics. Work on a copy.
- Enter header fields first, then one music row. Check barlines and durations before adding lyrics.
- Attach octave, duration, accidental and ornament suffixes to the intended note. Distinguish note grace brackets from barline ending brackets.
- Add voices and lyric verses one at a time; check alignment after each change.
- Use ~/^, bracket-height + and page settings deliberately to fix grouping/overlap. They do not repair wrong pitches.
- Compare each transcribed page with its original, including page transitions and annotations.
- Resolve diagnostics, save the editable master, export and inspect the final files.

27. Troubleshooting and feedback

Symptom	What to check
Unexpected symbols or warnings	ASCII punctuation, colon after the row prefix, balanced quotes/brackets, correct suffix attachment and lowercase keywords.
Lyrics shifted	C/Q voice pairing, @ for skipped notes, ~ for grouped Chinese text, / between English syllables.
Slur/ending overlaps	Correct endpoints, page boundary, manual + offsets and available vertical spacing.
Import fails	Supported file type, one input at a time, 100 MiB/200-page limits, readable unencrypted file and no active conflicting operation.
Transcription unavailable/fails	Packaged OCR worker, original input quality and Poppler availability for PDF processing.
Different font appearance	Actual installed OS family, bundled fallback, missing-glyph backup and external SVG viewer fonts.
Browser or update failure	Default-browser association and network; unavailable release information is not proof of being up to date.

For a reproducible issue, share the app version, OS, minimal JPS, relevant settings, steps and expected/actual result. Include original/reference images only when appropriate. Use Report Issues; proposed changes use the pull-request page. The historical specification's old personal contact details are not the OctoPus support route.

28. Reference basis and compatibility

This manual consolidates the locally retained Jianpu-Spec v1.0 chapters: introduction, descriptive header, music, lyrics and feedback. Image-only symbol/dynamic/barline tables were read and rewritten as text tables above. It also uses the captured Website-Contents/Zhipu-jsCode-editor for source formatting and source/preview-position conventions, checked against OctoPus behavior. The retained Zhipu-draw file is empty and contributes no additional instructions.

The notation syntax is inherited; the product/UI instructions and roadmap describe OctoPus. Historical claims about MIDI, playback, website auto-format settings or personal support contacts are not current desktop features. Runtime/parser, font policy, import limits and sample documents provide the current implementation cross-check. This is a rewritten usage guide, not a copy of the original scripts. References and authoring tools stay in OctoPus-dev; the production app ships this self-contained guide and its PDFs.

[Project home](#) • [Production README](#) • [Build environment](#)